

American International University- Bangladesh

Department of Computer Engineering

COE 3201: Data Communication Laboratory

Title: Study of Amplitude Modulator and Demodulator using Simulink

Abstract:

This experiment is designed to-

- 1.To understand the use of Simulink for AM modulation.
- 2.To develop understanding of AM demodulation.

Introduction:

Amplitude modulation (AM) is a one of the conventional technique used to transmit message signals using a carrier wave. The amplitude or strength of the high frequency carrier wave is modified in accordance with amplitude of the message signal.

AM Modulation:

Let us consider $m(t)$ is the message signal, and $c(t)$ is the high frequency carrier signal:

$$m(t) = A_m \sin(\omega_m t) = A_m \sin(2\pi f_m t)$$

$$c(t) = A_c \sin(\omega_c t) = A_c \sin(2\pi f_c t)$$

Then, AM modulated signal can be constructed as follows:

$$\begin{aligned} am(t) &= [A_c + m(t)] \sin(\omega_c t) = [A_c + A_m \sin(\omega_m t)] \sin(\omega_c t) \\ &= \left[A_c \left\{ 1 + \frac{A_m}{A_c} \sin(\omega_m t) \right\} \right] \sin(\omega_c t) = [A_c \{1 + \mu \sin(\omega_m t)\}] \sin(\omega_c t) \\ &= A_c \sin(\omega_c t) + \mu A_c \sin(\omega_c t) \sin(\omega_m t) \\ &= A_c \sin(\omega_c t) + \frac{\mu A_c}{2} [\cos\{(\omega_c - \omega_m)t\} - \cos\{(\omega_c + \omega_m)t\}] \end{aligned}$$

Here, $\mu = \frac{A_m}{A_c}$ known as AM modulation index. The value of $0 \leq \mu \leq 1$. Thus, $A_m < A_c$.

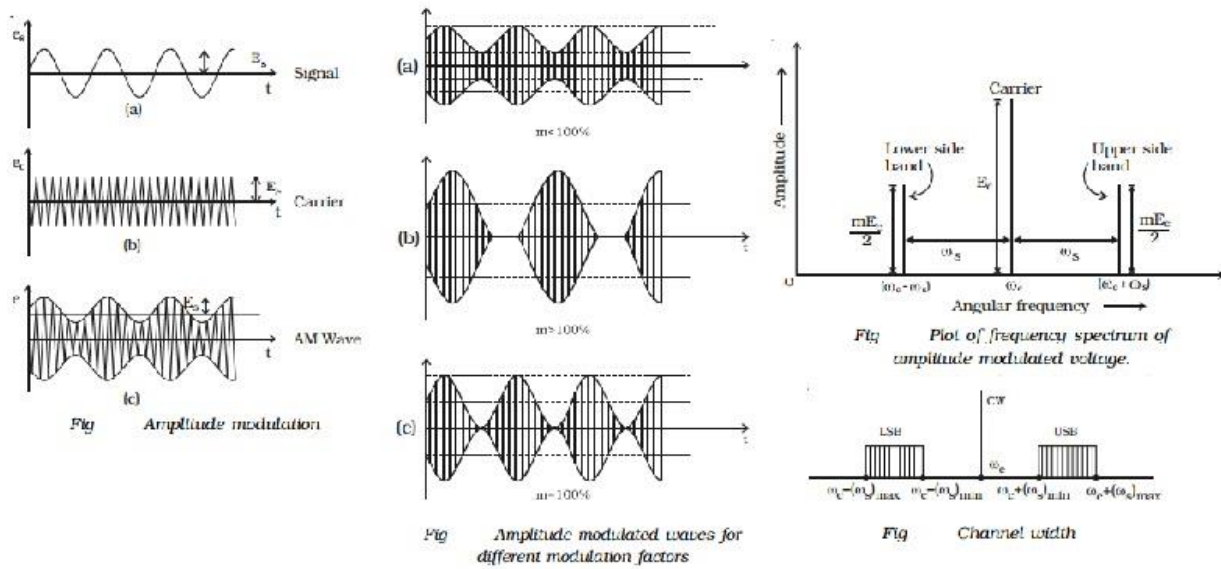


Fig. 1. AM modulated signal in time and frequency domain.

AM demodulation:

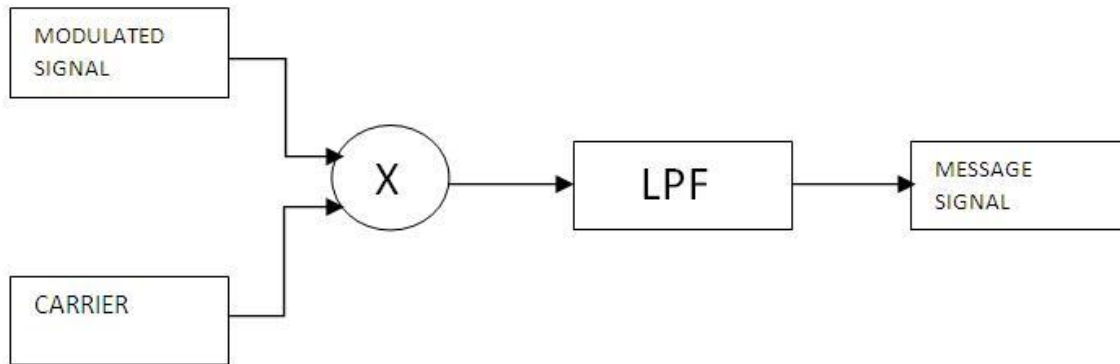


Fig. 2. AM demodulation process.

Thus, if we multiply the AM modulated signal with the carrier signal again in the demodulation process, one component becomes higher frequency $2\omega_c$ and other component becomes the low frequency baseband/modulating signal frequency ω_m as shown below:

$$\begin{aligned}
 A_c \cos\{(\omega_c - \omega_m)t\} \sin(\omega_c t) &= \frac{A_c}{2} [\sin\{(\omega_c - \omega_m + \omega_c)t\} - \sin\{(\omega_c - \omega_m - \omega_c)t\}] \\
 &= \frac{A_c}{2} [\sin\{(2\omega_c - \omega_m)t\} + \sin\{(\omega_m)t\}]
 \end{aligned}$$

As a result, if we use a low pass filter to remove the higher frequency component, then we can recover the low frequency baseband signal in the demodulation process.

MATLAB code:

```
clc;
clear;
fs=3000;
t = 0:1/fs:1-1/fs;
fm=5; %frequency of your modulating signal/baseband signal (original information
signal frequency)
fc=100; % frequency of your carrier signal, we will use for AM modulation

% Baseband Signal (Modulating Signal)
Am=10; % amplitude of your baseband signal
m = Am.*sin(2*pi*fm*t);

% Carrier Signal
Ac=20; % amplitude of your carrier signal
c= Ac.*sin(2*pi*fc*t);

% modulation index
mu=Am/Ac; % mu=10/20=0.5

% AM Modulated Signal
cam=(Ac.*(1+mu*sin(2*pi*fm*t))).*sin(2*pi*fc*t);

fftSignal = fft(cam); % This is frequency response of amplitude modulated signal
(cam).
fftSignal = fftshift(fftSignal)/(fs/2);
f = fs/2*linspace(-1,1,fs);

% AM demodulation part
% am_demodulated = abs(hilbert(cam))-20;

% multiplying the am modulated signal with carrier signal
am_demodulated =cam.*c;
% Applying Low-Pass filter
[k,l] = butter(6,(100).*(2/fs));
filtered_signal = filtfilt(k,l,(am_demodulated-200)./10) ;

% Plot the signals
figure;
subplot(4,1,1);
plot(t,m);
ylabel('amplitude');xlabel('time');
title('Modulating/Baseband Signal');
grid on

subplot(4,1,2);
plot(t,c);
ylabel('amplitude');xlabel('time');
title('Carrier Signal');
grid on

subplot(4,1,3);
plot(t,cam);
ylabel('amplitude');xlabel('time');
```

```

title('Amplitude Modulated Signal');
xlabel('Time (s)');
grid on

subplot(4,1,4);
%plot(t,am_demodulated);
plot(t,filtered_signal);
ylabel('Amplitude');xlabel('time');
title('Demodulated signal');
grid on

figure;
plot(f, abs(fftSignal));
axis([-200 200 0 25]);
title('FFT of AM modulated signal');
xlabel('Frequency (Hz)');
ylabel('Amplitude');

```

Generating AM in Simulink

For generating AM we just have to implement the equation of AM in block level.

Blocks Required

Analyzing the equation we need,

1. Carrier Signal Source
2. Message Signal Source
3. Blocks for viewing the signals – Scope
4. Product Block
5. Summer Block
6. Constant Block

We can find these blocks in the following locations of Simulink Library...

Carrier, Message, Constant blocks

- Simulink → Sources → Sine wave
- Simulink → Sources → Constant

View Block

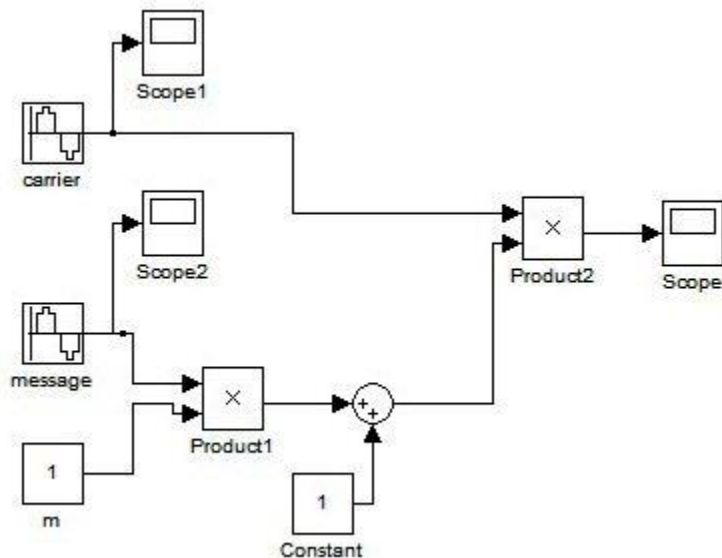
- Simulink → Sink → Scope

Product and Summer Block

- Simulink → Math Operations → Product

- Simulink -> Math Operations-> Summer

Block Diagram



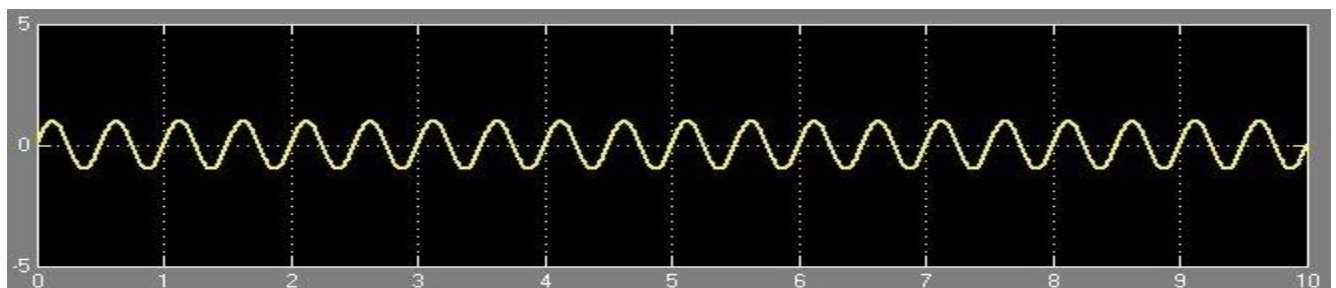
BLO

AM Generation using Simulink – Block Diagram

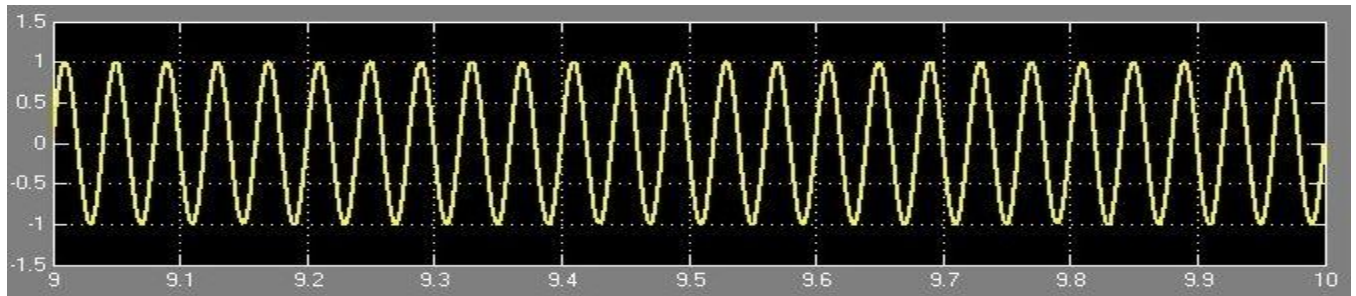
Block parameters can be changed by selecting the block and parameter:

- Carrier Signal frequency = $2\pi \cdot 25$ and sampling time = $1/5000$
- Message Signal frequency = 2π and sampling time = $1/5000$
- Amplitudes of both signals are 1

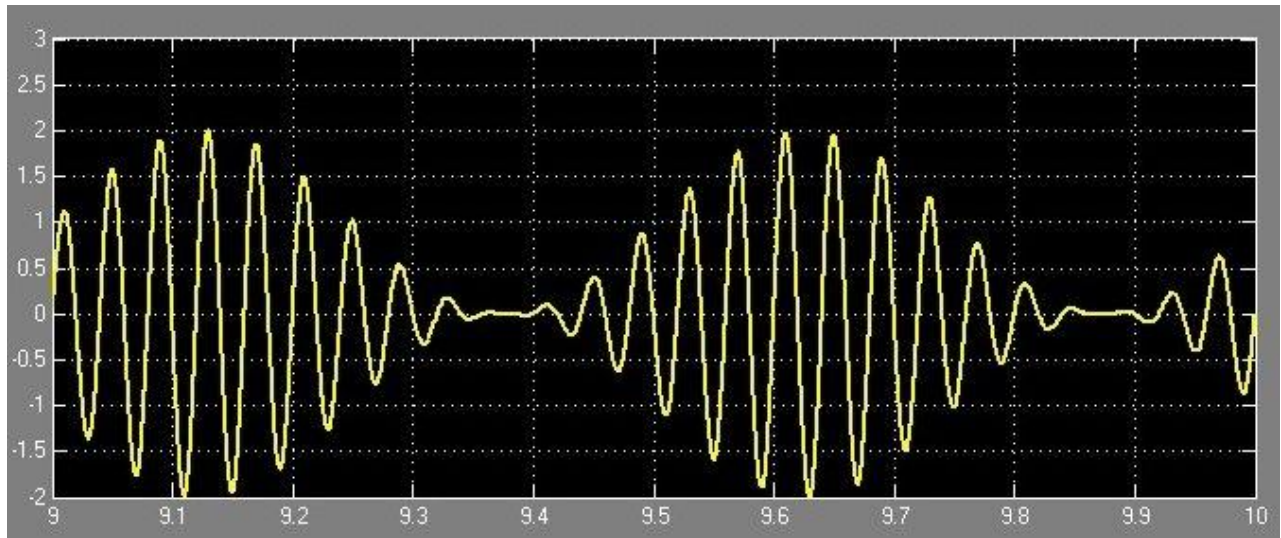
Output Waveforms



AM Generation using Simulink – Message Signal



AM Generation using Simulink – Carrier



AM Generation using Simulink – Modulated Signal

Performance Task:

Implement the following demodulation in Simulink to retrieve the original signal:

You have a signal ' $m(t) = (2*\sin(2*\pi*4*t)+3*\cos(2*\pi*6*t))$ '. Apply amplitude modulation (AM) on the given signal with carrier signal ' $c(t) = \cos(2*\pi*50*t)$ ', and then do demodulation to recover the original message signal $m(t)$.

Implement the following demodulation in Simulink to retrieve the original signal:

